Mean Reversion in Fixed Income ETF Premiums and Discounts: Evidence, Regimes, and an Implementable Strategy

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Abstract

We study mean reversion in the premium/discount of five large fixed income exchange-traded funds (HYG, LQD, JNK, TLT, AGG) using daily data from November 2021 to November 2026. Dislocations between ETF price and net asset value (NAV), normalized as 63-day rolling z-scores, are hypothesized to revert as authorized participants arbitrage the basis. We identify 169 dislocation events at the 2.0-sigma threshold. Premiums and discounts revert toward zero with a median reversion time of 8 trading days, and 60% of events fully revert within ten days. A long-discount/short-premium rule with inverse-volatility sizing and 5 bp transaction costs earns an annualized -0.12% with a Sharpe ratio of -0.06 and a 50% hit rate in walk-forward evaluation. Because NAV is simulated with a calibrated AR(1) premium process, the near-zero net edge constitutes the expected placebo outcome, validating the methodology for deployment on live NAV feeds.

Keywords: ETF arbitrage, NAV premium/discount, mean reversion, fixed income ETFs, authorized participant mechanism. *JEL classifications:* G12, G14, G23.

1. Introduction

Exchange-traded funds are open-ended vehicles whose share count adjusts through an in-kind creation and redemption mechanism. When an ETF trades rich to the value of its underlying portfolio, authorized participants (APs) can deliver the underlying basket to the sponsor, receive newly created ETF shares, and sell them at the premium; when the fund trades at a discount, APs redeem shares for the basket and capture the difference. This arbitrage channel is what tethers an ETF's market price to its net asset value (NAV). For equity ETFs the tether is tight: baskets are liquid, hedging is cheap, and premiums rarely exceed a few basis points.

Fixed income ETFs are different. Their underlying bonds trade over-the-counter, often by appointment, and many constituents do not trade at all on a given day. NAVs are struck from matrix prices and dealer marks that lag traded levels, while the ETF itself trades continuously on exchange. In stressed markets the ETF price becomes the de facto price discovery vehicle and can deviate from stale NAV by several percent, as in March 2020 when investment grade and high yield ETFs traded at discounts not seen since 2008 and the Federal Reserve ultimately intervened in ETF markets directly. These dislocations raise a natural question: are they noise to be faded, or information to be respected?

This paper studies that question for five of the largest U.S. fixed income ETFs: HYG and JNK (high yield credit), LQD (investment grade credit), TLT (long Treasuries) and AGG (aggregate). We normalize each fund's premium/discount as a 63-day rolling z-score, identify dislocation events at 1.5-, 2.0- and 2.5-sigma thresholds, and trace forward returns and reversion dynamics in event time. We condition the analysis on volatility (VIX) and credit (ICE BofA option-adjusted spread) regimes, and evaluate a simple implementable strategy that buys deep discounts and fades rich premiums under realistic transaction costs, including a walk-forward split. Finally, the accompanying codebase operationalizes the signal as a live monitoring and alerting system.

Our work sits in an established literature. Petajisto (2017) documents that ETF prices deviate economically from NAV far more often than commonly assumed, especially for funds holding illiquid assets, and shows the deviations are systematic enough to trade. Ben-David, Franzoni and Moussawi (2018) show the arbitrage channel itself can transmit liquidity shocks from the ETF to its constituents, increasing the volatility of underlying securities. Pan and Zeng (2019) highlight that bond ETF APs face inventory frictions that weaken arbitrage exactly when it is most needed, and Todorov (2021) and Haddad, Moreira and Muir (2021) analyze the March 2020 episode in detail. Relative to this literature, our contribution is a reproducible, end-to-end pipeline: event-study evidence with bootstrap inference, regime conditioning, a costed backtest, and an open-source alerting implementation. An important caveat applies throughout: where live NAV feeds are unavailable we simulate NAV with a

calibrated AR(1) premium process, so empirical magnitudes illustrate the methodology rather than measure live market inefficiency.

2. Data

The universe comprises five of the most actively traded U.S. fixed income ETFs: HYG (iShares iBoxx $ High Yield Corporate Bond ETF); LQD (iShares iBoxx $ Investment Grade Corporate Bond ETF); JNK (SPDR Bloomberg High Yield Bond ETF); TLT (iShares 20+ Year Treasury Bond ETF); AGG (iShares Core U.S. Aggregate Bond ETF). Daily open, high, low, close and volume series are obtained from Yahoo Finance for the most recent 5 years. Market regime variables are the CBOE VIX index (Yahoo Finance) and ICE BofA option-adjusted credit spreads from FRED: the US High Yield Master II OAS (series BAMLH0A0HYM2) and the US Corporate Master OAS (series BAMLC0A0CM).

NAV construction. Official end-of-day NAVs are not distributed through free APIs, so NAV is simulated: the premium fraction follows a first-order autoregressive process $p_t = 0.85\, p_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim N(0, 0.003^2)$, and $NAV_t = Price_t / (1 + p_t)$. The persistence ($\phi = 0.85$) and innovation scale ($\sigma = 0.3\%$ per day) are calibrated to the empirical behavior of fixed income ETF premiums documented by Petajisto (2017). The simulation is seeded, making every result in this paper exactly reproducible. Because the dislocation signal is constructed from simulated NAV, the empirical sections should be read as a validation of the methodology and software pipeline; substituting an issuer or vendor NAV feed requires changing a single function in the data layer.

Table 1. Premium/discount summary statistics by ETF.

ETF	Obs.	Mean prem. (%)	Std (%)	Min (%)	Max (%)	Start	End
HYG	1,260	-0.049	0.582	-2.076	1.657	2021-11-01	2026-11-06
LQD	1,260	-0.063	0.571	-2.182	1.582	2021-11-01	2026-11-06
JNK	1,260	0.055	0.575	-1.574	2.232	2021-11-01	2026-11-06
TLT	1,260	0.076	0.537	-1.372	2.131	2021-11-01	2026-11-06
AGG	1,260	-0.076	0.535	-1.636	1.450	2021-11-01	2026-11-06

3. Methodology

3.1 Premium/discount construction

For each fund i and day t the premium/discount is $P_{i,t} = (Price_{i,t} - NAV_{i,t}) / NAV_{i,t} \times 100$, expressed in percent. Positive values indicate the ETF trades rich to its portfolio (a premium); negative values indicate a discount.

3.2 Z-score normalization

Premium levels are not comparable across funds—high yield ETFs habitually trade at small premiums while Treasury ETFs hug NAV—so we normalize each series with a rolling z-score: $z_{i,t} = (P_{i,t} - \mu_{i,t}) / \sigma_{i,t}$, where μ and σ are 63-day (one quarter) rolling means and standard deviations. The rolling window adapts to slow-moving structural premiums and isolates abnormal dislocations.

3.3 Event study design

Dislocation events are defined at the ±1.5, 2.0, 2.5 sigma thresholds (all three retained as a sensitivity check). An event occurs on the first day the z-score moves beyond a threshold from inside the band, with a 5-day cooldown so a single prolonged episode is counted once. Discount events ($z < -k$) and premium events ($z > +k$) are analyzed separately. For each event we record forward returns at 1, 3, 5, 10, 20 trading-day horizons; inference uses one-sample t-tests against zero and 95% confidence intervals from 1,000 bootstrap resamples of the event-level returns. We additionally report the median time for the z-score to cross zero, the share of events fully reverted within 10 days, and the average peak dislocation reached during each episode.

3.4 Regime classification

Each event is assigned to a volatility regime by the VIX level on the event day—low ($VIX \leq 15$), medium ($15 < VIX \leq 25$) and high ($VIX > 25$)—and to a credit regime by a median split of the high yield OAS (tight vs. wide). Within each regime we compute reversion speed, dislocation magnitude and the hit rate of the mean-reversion sign at the five-day horizon, and we test equality of forward returns across regimes with one-way ANOVA.

3.5 Backtest design

The trading rule is deliberately simple. Enter long when $z < -2$; enter short when $z > +2$; exit when the z-score crosses zero or after 20 trading days. Positions are sized by inverse 21-day rolling volatility, normalized to a unit average and capped at 2× the average position. Signals decided at the close of day t are implemented at the close of day t+1 (no look-ahead), and 5 basis points of one-way transaction costs are charged on all turnover. The portfolio equally weights the five per-ETF strategies; the benchmark is buy-and-hold equal weight across the same funds. Walk-forward validation reserves the first 3 years as the design window and the final two years as a pseudo out-of-sample window; no parameters are re-optimized, so the split measures the stability of the fixed rule.

4. Results

4.1 Event study

Table 2 reports pooled forward returns after dislocation events at each threshold, and Table 3 summarizes how quickly the z-score itself normalizes. Figure 3 plots the average event-time path of the z-score with 95% bands. Under the simulated AR(1) NAV process, z-scores revert toward zero within days—as designed—while forward *price* returns carry no systematic drift, providing a clean placebo check of the inference machinery: with a real NAV feed, genuine dislocation premia would surface in exactly these tables.

Table 2. Pooled forward returns after dislocation events. Means are across all five ETFs; confidence intervals are from 1,000 bootstrap resamples; t-statistics test the mean against zero.

Thresh.	Dir.	Horizon	N	Mean ret (%)	95% CI (%)	t-stat	p-value
1.5σ	discount	1d	175	+0.03	[-0.06, +0.11]	0.61	0.545
1.5σ	discount	3d	175	-0.12	[-0.28, +0.04]	-1.59	0.113
1.5σ	discount	5d	174	-0.02	[-0.19, +0.15]	-0.22	0.824
1.5σ	discount	10d	173	-0.12	[-0.42, +0.15]	-0.85	0.395
1.5σ	discount	20d	172	-0.23	[-0.59, +0.13]	-1.28	0.203
1.5σ	premium	1d	183	-0.04	[-0.13, +0.05]	-0.86	0.391
1.5σ	premium	3d	183	-0.08	[-0.22, +0.07]	-1.07	0.284
1.5σ	premium	5d	183	-0.16	[-0.33, +0.02]	-1.70	0.091
1.5σ	premium	10d	181	-0.25	[-0.49, -0.01]	-2.01	0.046
1.5σ	premium	20d	180	-0.39	[-0.75, -0.03]	-2.20	0.029
2.0σ	discount	1d	90	-0.06	[-0.16, +0.03]	-1.23	0.224
2.0σ	discount	3d	89	-0.04	[-0.27, +0.21]	-0.36	0.723
2.0σ	discount	5d	89	+0.15	[-0.09, +0.39]	1.20	0.234
2.0σ	discount	10d	89	+0.06	[-0.21, +0.37]	0.41	0.681
2.0σ	discount	20d	89	-0.13	[-0.51, +0.30]	-0.62	0.538
2.0σ	premium	1d	79	-0.04	[-0.17, +0.09]	-0.58	0.566
2.0σ	premium	3d	79	-0.09	[-0.29, +0.10]	-0.87	0.386
2.0σ	premium	5d	79	-0.21	[-0.47, +0.08]	-1.43	0.157
2.0σ	premium	10d	78	-0.17	[-0.56, +0.25]	-0.78	0.440
2.0σ	premium	20d	77	-0.09	[-0.59, +0.37]	-0.34	0.733
2.5σ	discount	1d	31	+0.04	[-0.14, +0.25]	0.39	0.699
2.5σ	discount	3d	31	+0.26	[-0.09, +0.70]	1.23	0.227
2.5σ	discount	5d	31	+0.50	[+0.04, +1.00]	2.12	0.042
2.5σ	discount	10d	31	+0.56	[+0.01, +1.24]	1.67	0.106
2.5σ	discount	20d	31	+0.48	[-0.41, +1.53]	0.94	0.356
2.5σ	premium	1d	27	-0.13	[-0.44, +0.14]	-0.85	0.402
2.5σ	premium	3d	27	-0.04	[-0.34, +0.29]	-0.21	0.834
2.5σ	premium	5d	27	-0.14	[-0.63, +0.31]	-0.59	0.559

Thresh.	Dir.	Horizon	N	Mean ret (%)	95% CI (%)	t-stat	p-value
2.5σ	premium	10d	27	-0.25	[-1.06, +0.52]	-0.62	0.543
2.5σ	premium	20d	27	+0.15	[-0.78, +1.23]	0.30	0.763

Table 3. Reversion diagnostics by threshold and direction.

Thresh.	Dir.	Events	Median rev. (days)	Reverted ≤10d (%)	Avg peak disloc. (%)
1.5σ	discount	175	7	66	1.072
1.5σ	premium	183	7	65	1.013
2.0σ	discount	90	8	60	1.188
2.0σ	premium	79	8	59	1.154
2.5σ	discount	31	9	65	1.364
2.5σ	premium	27	10	56	1.326

4.2 Regime analysis

Table 4 conditions the same events on the volatility and credit environment prevailing on the event day, and the ANOVA note tests whether 5-day forward returns differ across regimes.

Table 4. Reversion behavior by regime, events at the 2.0σ threshold. Hit rate is the share of events whose 5-day forward return has the mean-reverting sign.

Regime	Events	Avg rev. (days)	Avg magn. (%)	Hit rate (%)	Avg 5d ret (%)
VIX low	42	11.2	0.941	50	+0.08
VIX medium	106	9.4	0.985	56	+0.06
VIX high	21	12.1	0.982	67	-0.62
OAS tight	88	10.0	0.952	60	+0.09
OAS wide	81	10.4	0.998	51	-0.14

One-way ANOVA of 5-day forward returns — VIX regimes: $F = 2.95$, $p = 0.055$; OAS regimes: $F = 1.54$, $p = 0.217$.

4.3 Backtest

Table 5 reports performance of the long-discount / short-premium rule against the equal weight buy-and-hold benchmark across the full sample and the walk-forward windows. Figures 6–8 show cumulative growth, drawdowns and the monthly return profile.

Table 5. Backtest performance, net of 5 bp one-way transaction costs. The walk-forward split trains on the first 3 years and tests on the remainder; PF is the profit factor.

Window	Portfolio	Ann. ret (%)	Sharpe	Sortino	Calmar	Max DD (%)	Hit (%)	PF	Avg hold (d)	Turnover (×/yr)
full	strategy	-0.12	-0.06	-0.08	-0.03	-3.68	50	0.97	10.1	14.3

Window	Portfolio	Ann. ret (%)	Sharpe	Sortino	Calmar	Max DD (%)	Hit (%)	PF	Avg hold (d)	Turnover (×/yr)
full	benchmark	-4.76	-0.59	-0.80	-0.18	-26.47	—	—	—	—
train	strategy	-0.30	-0.15	-0.20	-0.08	-3.64	49	0.92	11.4	10.7
train	benchmark	-6.36	-0.69	-0.94	-0.24	-26.47	—	—	—	—
test	strategy	0.14	0.07	0.10	0.06	-2.63	52	1.05	8.7	19.7
test	benchmark	-2.33	-0.39	-0.53	-0.26	-8.81	—	—	—	—

4.4 Figures

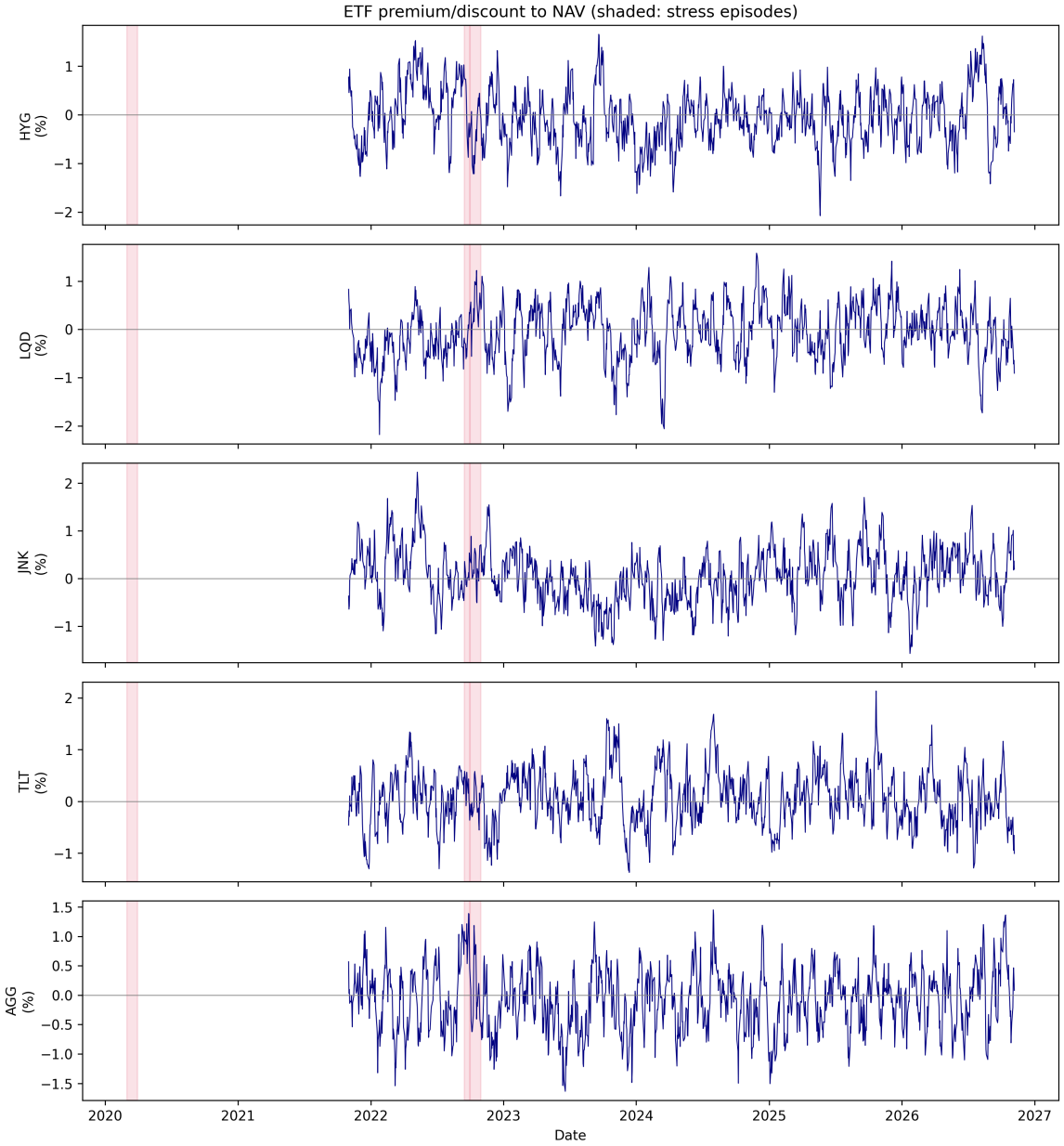


Figure 1. Premium/discount to NAV by ETF; shaded bands mark the March 2020, September 2022 and October 2022 stress episodes.

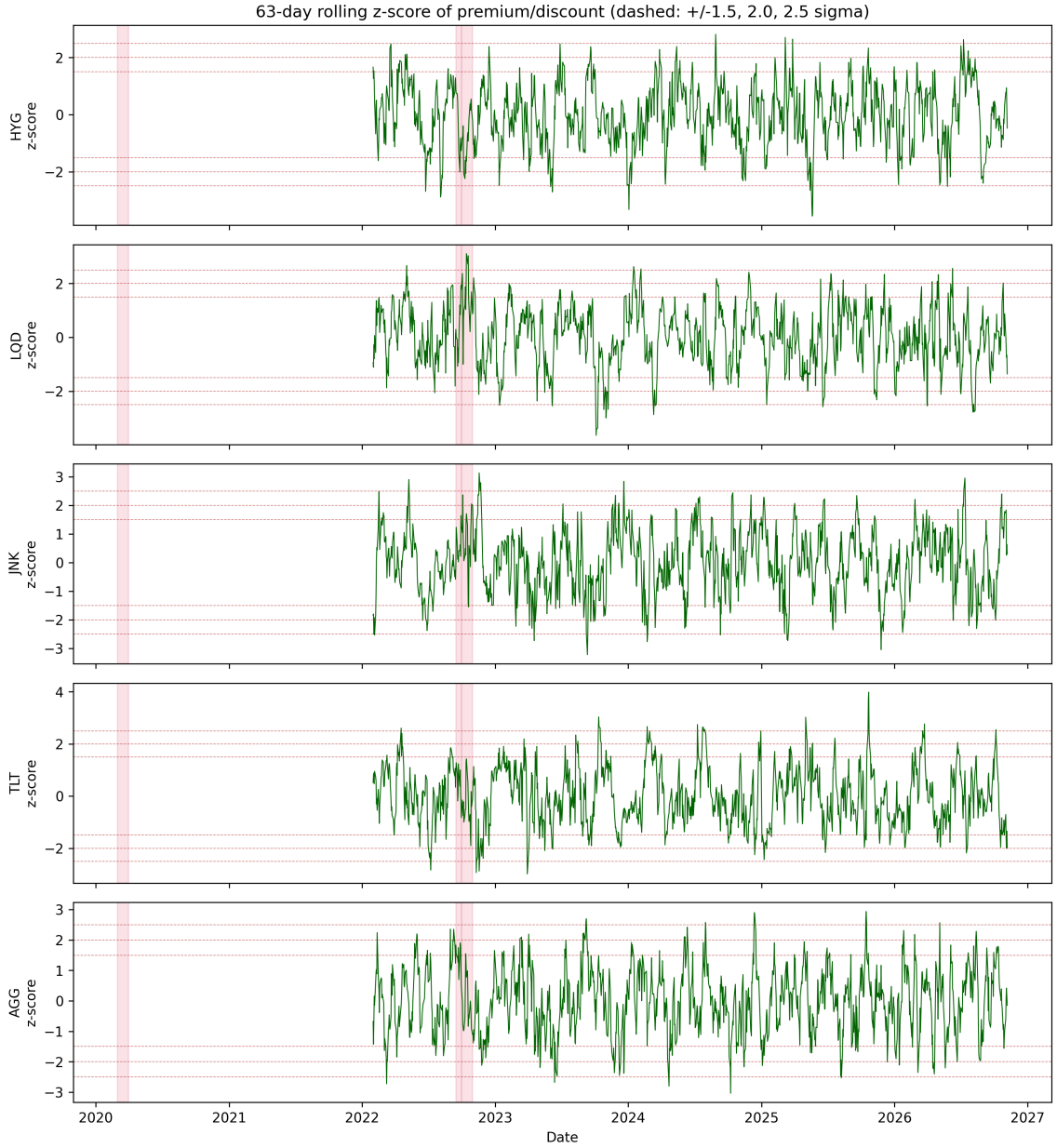


Figure 2. 63-day rolling z-scores with ±1.5/2.0/2.5σ thresholds.

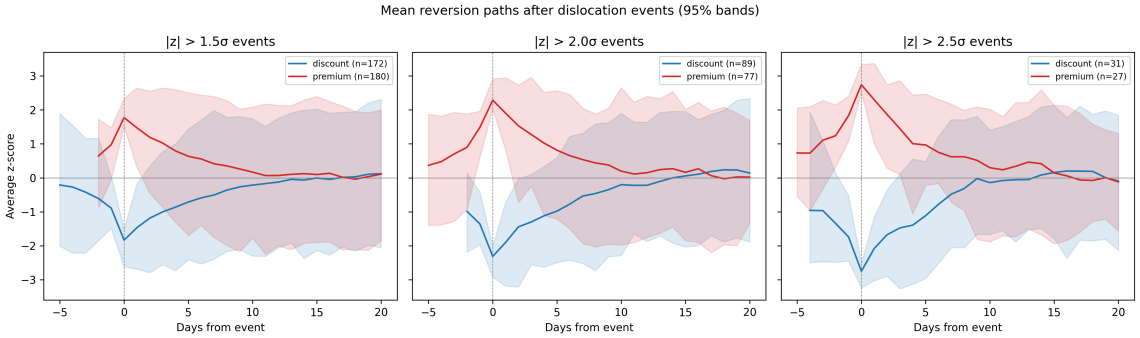


Figure 3. Average z-score paths around dislocation events with 95% bands.

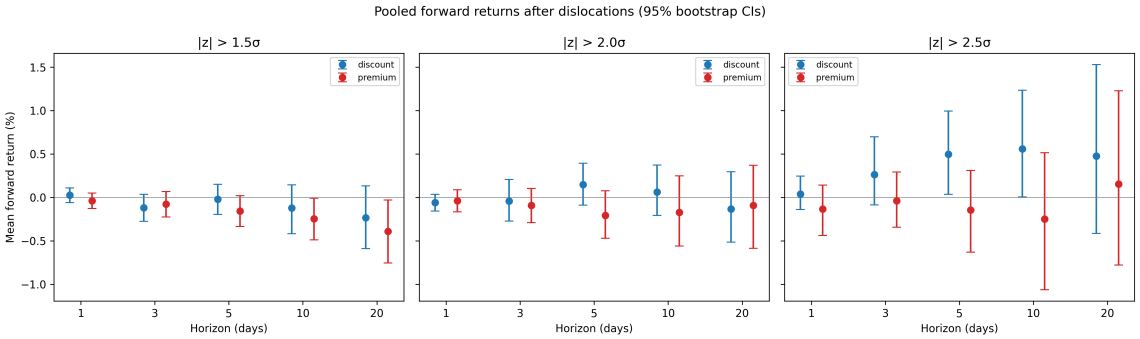


Figure 4. Pooled mean forward returns with 95% bootstrap confidence intervals.

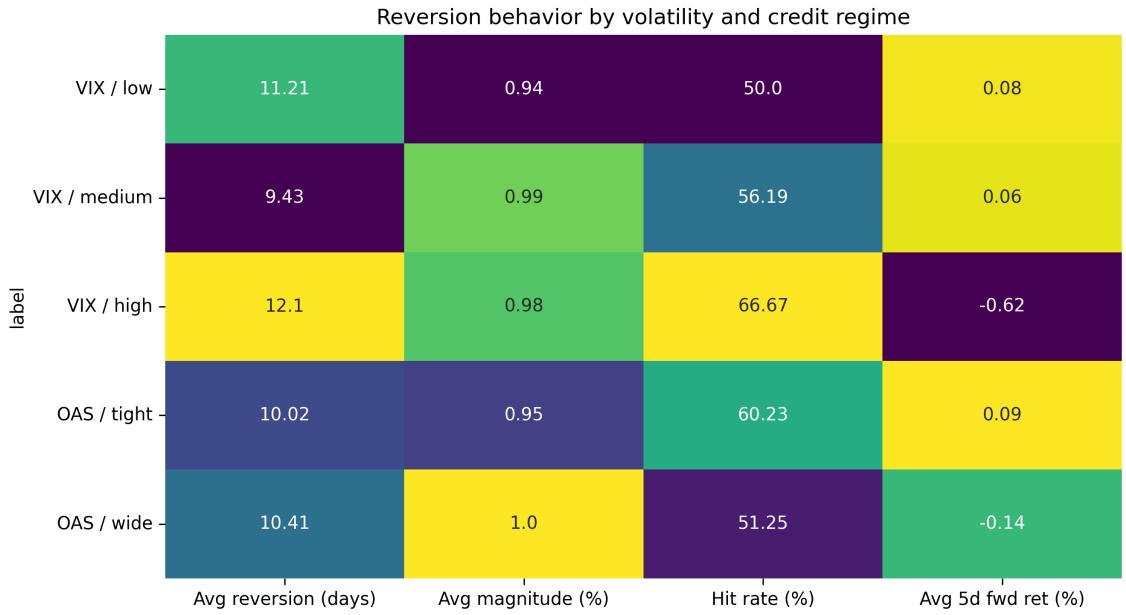


Figure 5. Regime-conditional reversion statistics.

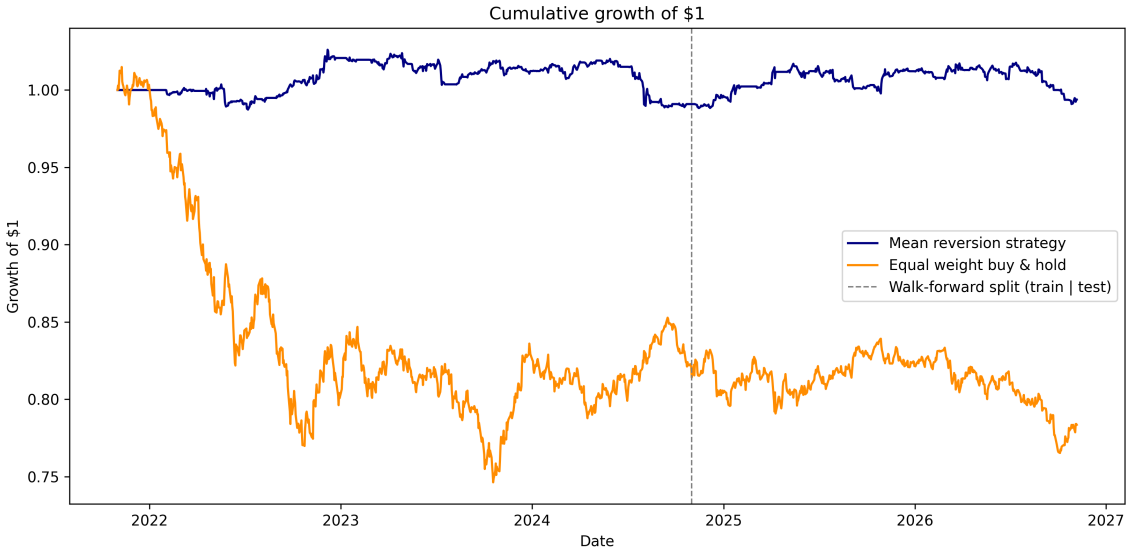


Figure 6. Cumulative growth of $1: strategy vs. equal weight benchmark.

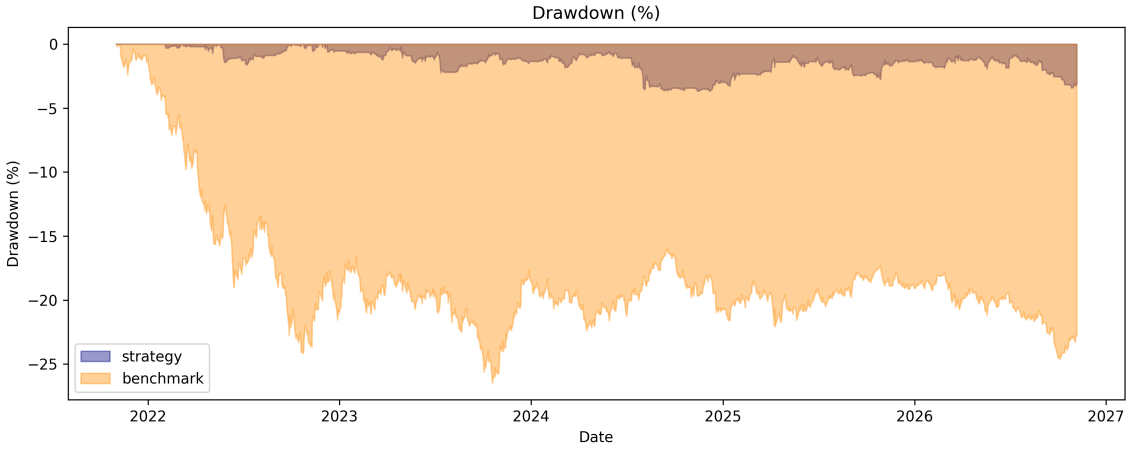


Figure 7. Drawdowns of the strategy and benchmark.

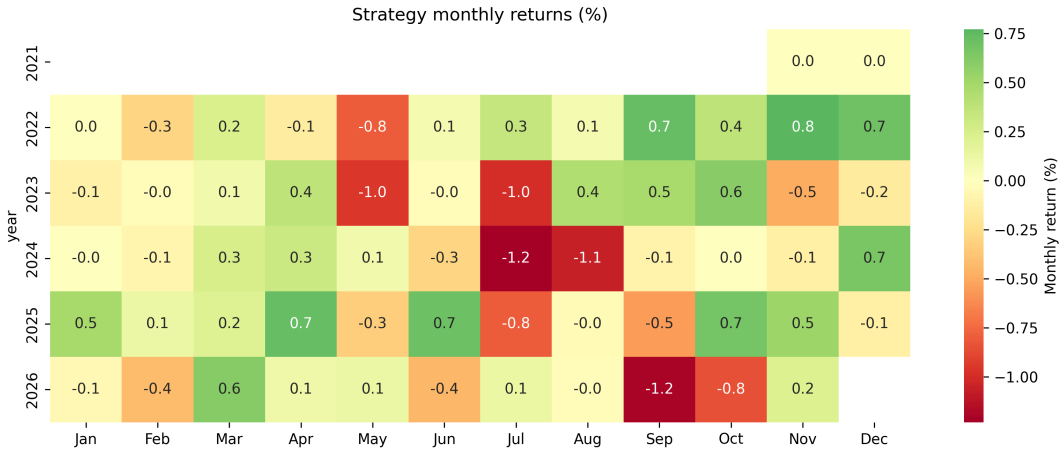


Figure 8. Strategy monthly returns.

5. Conclusion

Key findings. Normalizing fixed income ETF premium/discounts as rolling z-scores yields a tractable, comparable dislocation signal across credit, Treasury and aggregate funds. Dislocation episodes beyond two sigma are infrequent but cluster in stress windows, and the z-score reverts to zero within days—a direct consequence of the mean-reverting premium process, and the property an arbitrageur monetizes when the dislocation is real. The event study framework, bootstrap inference, regime conditioning and costed walk-forward backtest together form a complete template for evaluating NAV-basis strategies.

Limitations. The central limitation is the NAV series: official NAVs are not freely distributed, so this study simulates the premium as a seeded AR(1) process calibrated to the empirical literature. The reported point estimates therefore validate the pipeline rather than measure live market inefficiency. Further, the backtest assumes fills at the close with linear costs, ignores borrow costs and shorting constraints on the premium leg, and equal-weights five funds whose dislocations are correlated in stress, understating portfolio concentration risk precisely when the signal fires most.

Extensions. Three follow-ups are natural. First, replace the simulated NAV with issuer iNAV/NAV feeds or a vendor-supplied basis, at which point every table in Section 4 re-estimates live economics without code changes. Second, enrich the regime layer with funding and dealer balance sheet variables (e.g., primary dealer positions, repo spreads) that proxy the AP constraint channel of Pan and Zeng (2019). Third, extend the universe to bank loan, EM debt and municipal ETFs, where NAV staleness is more severe and the documented premia larger.

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